

m-format Constellation UltraQuickStart

DISCLAIMER — UNTESTED ALPHA SOFTWARE

This software has not yet been independently tested, audited, or proven in production. Do not use the m-format constellation to back up significant sums of money at this time. Doing so is tantamount to asking to be rekt. Use only with disposable amounts (a few sats), on testnet, or for evaluation. The codecs, CLIs, BCH error-correcting math, BIP-388 wallet-policy emission, and cross-card binding invariants have all been authored and reviewed only by the original developer; assume bugs until external review happens.

m-format Constellation UltraQuickStart

A two-page 3-card steel-backup recipe for a single-sig Bitcoin wallet. For the gentle ~42-page newcomer walkthrough see `m-format-quickstart.pdf`; for the comprehensive 121-page reference see `m-format-manual.pdf`.

Install

```
cargo install --locked --git https://github.com/bg002h/mnemonic-toolkit
mnemonic --version
```

Bundle

Generate fresh entropy on an air-gapped machine for a real wallet. This recipe uses the **public BIP-39 test phrase** so the output is reproducible — never engrave or fund a wallet derived from it.

```
mnemonic bundle \
  --network mainnet \
  --template bip84 \
  --slot @0.phrase="abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aba
```

Output: three card strings (msl..., mk1..., md1...) plus a chunked-form engraving guide. Each card has its own BCH error-correcting checksum.

Verify (before stamping)

```
mnemonic verify-bundle \
  --network mainnet --template bip84 \
  --slot @0.phrase="abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aba
  --msl <msl-string> \
  --mk1 <mk1-line-1> --mk1 <mk1-line-2> \
  --md1 <md1-line-1> --md1 <md1-line-2> --md1 <md1-line-3>
```

result: ok → safe to stamp. A *_decode: error at position N line names the single character whose checksum disagrees — re-stamp character N of the failing card.

Stamp + separate

Stamp all three card strings on rated steel, re-decode the engraved set with `verify-bundle` against the steel-read strings, then geographically separate the three plates:

- **Red** (msl) — primary safe.
- **Blue** (mk1) — bank safe-deposit box.
- **Green** (md1) — off-site (trusted person, attorney, vault).

The split-recovery property: an attacker who recovers only one plate cannot derive the wallet alone.

Recover

Read the cards off steel and reverse:

```
mnemonic convert --from msl=<msl-string> --to phrase
mnemonic convert --from mk1=<mk1-string> --to xpub --to fingerprint --to path
md decode <md1-line-1> <md1-line-2> <md1-line-3>
```

`mnemonic convert` recovers the seed phrase from `ms1`, the `xpub` + origin from `mk1`. `md decode` takes the three `md1` strings positionally and returns the wallet policy.

For `multisig` (`--template wsh-sortedmulti --threshold 2 --slot @0.phrase=... --slot @1.phrase=... --slot @2.phrase=...`) or `watch-only` (`--slot @0.xpub=...`), see the QuickStart's Parts III + IV.